EE4-55
EE9-AO11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

Corrected Copy

MEMS AND NANOTECHNOLOGY

Monday, 20 May 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question 1.
Answer Question 2 OR Question 3.
Answer Question 4 OR Question 5.

Question 1 carries 40% of the marks. Remaining questions carry 30% each.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	Z. Durrani, A.S. Holmes, Z. Durrani
	Second Marker(s) :	A.S. Holmes, Z. Durrani, A.S. Holmes

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Information for Candidates

The permittivity of free space $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

The fractional resistance change for a <110>-aligned piezoresistor on a <100>-oriented silicon wafer, subject to longitudinal and transverse axial stresses σ_L, σ_T, can be expressed as:

$$\frac{\Delta R}{R} = \pi_L \sigma_L + \pi_T \sigma_T$$

where π_L and π_T are the reduced piezoelectric coefficients.

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This question is compulsory

1. a) Figure 1.1 shows the unit cell in a Si crystal, where each side $a = 0.54$ nm. Using this diagram, calculate the concentration of atoms in crystalline Si, per cubic metre. [4]

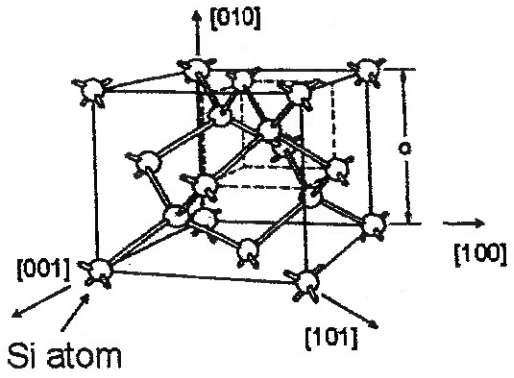


Figure 1.1

b) A Si (100) surface with an oxide layer of thickness $X_i = 10$ nm is to be further oxidised, at 1000 °C. At this temperature, the oxidation 'rate' constants in the Deal-Grove equation for oxide growth:

$$X^2 + AX = B(t + \tau)$$

are $A = 0.3 \times 10^4$ Å and $B = 2.9 \times 10^4$ Å^2/min, with X the thickness of the oxide layer.

Calculate the total thickness of the oxide grown under these conditions after 50 minutes. [5]

c) Consider an enhancement mode, n-channel, Si MOSFET, with gate oxide thickness t_{ox}, and energy separation $e\psi_B$ between the Fermi (E_F) and the intrinsic (E_i) levels. Here, e is the electron charge.

i) Sketch the energy band diagram (E_c, E_v, E_i, and E_F) along a line perpendicular to the gate oxide - bulk Si plane, for 'flat-band' conditions. Your diagram should show the energy bands in the bulk Si, oxide, and gate regions. [3]

ii) Sketch the energy band diagram (E_c, E_v, E_i, and E_F) along the line perpendicular to the gate oxide – bulk Si plane, for the following condition: there is a trapped charge $+Q_f$ at the gate oxide/Si interface, and the work function in the gate metal $\phi_m < \phi_s$, the work function in the Si. [3]

Question 1 continues on the next page.

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Question 1 continued.

d) Using suitable diagrams, explain the vapour-liquid-solid (VLS) growth mechanism for a silicon nanowire. [5]

e) Starting from the generalised form of Hooke's law, and applying appropriate constraints to the stress and strain components, derive an expression for the biaxial modulus of a material in terms of its Young's modulus and Poisson's ratio. Give an example of a situation where the biaxial modulus would be used in analysis of a MEMS device. [5]

f) Figure 1.2 shows a cross-section of a MEMS device with integrated electronics, fabricated by MEMS-first processing. Identify the materials labelled A to E. Also, draw a diagram showing how the cross-section would have appeared immediately before the electronics was added. What are the advantages of MEMS-first processing over other integration approaches? [5]

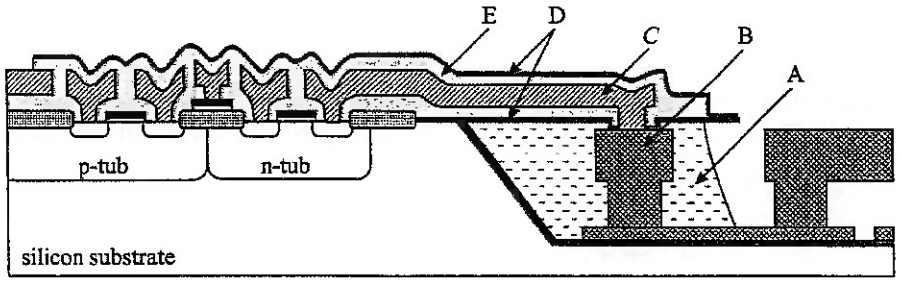


Figure 1.2

g) In a particular MEMS application, there is a need to align two silicon parts parallel to each other and at a well-defined spacing g. The approach taken is to use sapphire spacer balls located in truncated pyramidal pits formed by anisotropic etching, as shown in Figure 1.3. If the sapphire balls have a diameter of 500 μm, and the target spacing is $g = 200$ μm, what should be the side length of the pyramidal pits at the silicon top surface? Also, if the anisotropic etchant has a selectivity of 400, and the etching depth is controlled to give a clearance of $h = 25$ μm beneath each spacer ball, what size of mask aperture should be used for the anisotropic etching process? [5]

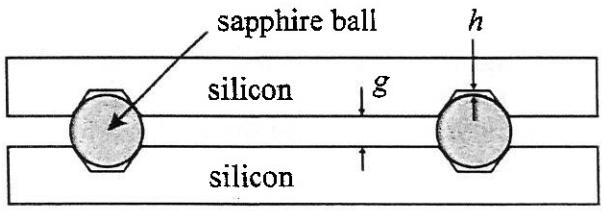


Figure 1.3

Question 1 continues on the next page.

Question 1 continued.

h) By considering the imaging of a periodic line/space pattern, show that the minimum resolvable half-pitch R for a projection lithography system with on-axis, plane wave illumination is limited to:

$$R = 0.5\frac{\lambda}{NA}$$

where λ is the optical wavelength and NA is the numerical aperture. [5]

End of Question 1.

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2. An electron travelling in the positive x-direction is incident from the region $x < 0$ onto a one-dimensional potential barrier of height V_0 and width L. The potential energy $V(x)$ is given by:

$$V(x) = V_0, \text{ for } 0 \leq x \leq L$$
$$V(x) = 0, \text{ elsewhere}$$

a) Write down the general form of the time-independent wavefunctions $\psi(x)$ when $x < 0$, x lies between 0 and L, and when $x > L$. Your answer should include expressions for the wave-vectors in these three regions of x. [8]

b) Using the wavefunctions $\psi(x)$ in the different regions, and the boundary conditions at $x = 0$ and $x = L$, write down equations linking the wavefunctions at $x = 0$ and $x = L$. [8]

c) Hence, derive an expression for the amplitude transmission coefficient t for the electron travelling across the barrier. [14]

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3. a) Figure 3.1 shows a one-dimensional (1-D), periodic potential, with potential wells of width b, depth V_0 and period a.

i) Write down the wavevectors α and β in Region I ($0 < x < a - b$) and Region II ($-b < x < 0$) respectively. Hence, write down expressions for the wave-functions ψ_I and ψ_{II} in Regions I and II respectively. [4]

ii) Find the Bloch functions u_I and u_{II}, associated with Regions I and II respectively, for this potential. Here, you may use Bloch's Theorem, which states that the wave-function in a 1-D periodic potential with period a is given by $\psi(x) = u(x).\exp(ikx)$, where $u(x)$ also has period a. [6]

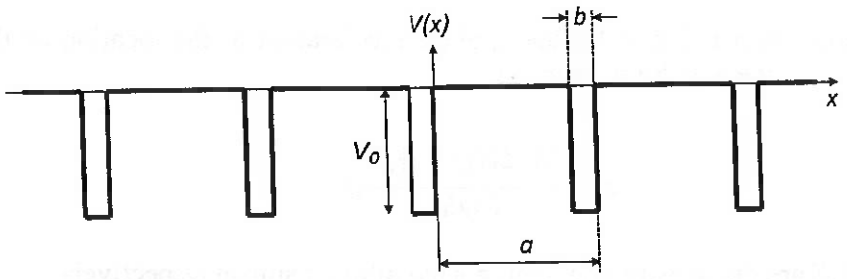


Figure 3.1

b) An ideal, one-dimension conductor of length L is placed between metallic source and drain regions. The Fermi energies of the source and drain regions are E_{F1} and E_{F2} respectively. If the one-dimensional density of states in k-space is given by $n(k) = 1/\pi$, show that the conductance of an electron channel along the conductor is given by $G = 2e^2/h$. [14]

c) Using suitable diagrams, explain briefly the operation of an Esaki diode. [6]

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4. a) What are the main transduction mechanisms used in MEMS sensors? Which two mechanisms are most widely used, and why are these preferred? [6]

b) Figure 4.1 shows a silicon micromachined accelerometer comprising a proof mass supported by two cantilevers. A p-type piezoresistive bridge located at the root of one of the cantilevers is used for readout. The dimensions of each cantilever are $l = 500$ μm, $w = 200$ μm, $h = 20$ μm, and those of the proof mass are $L = W = 2$ mm, $H = 400$ μm.

i) Write down an expression for the bending moment in either cantilever, as a function of position along the cantilever, when the accelerometer is subject to an upward acceleration a normal to the wafer. You should neglect gravity and assume the cantilevers are mass-less. [4]

ii) Show that the strain induced by the applied acceleration at the location of the piezoresistive bridge may be written as:

$$\varepsilon = \frac{3WLH(l + \frac{L}{2})\rho}{Ewh^2} a$$

where ρ and E are the density and Young's modulus of silicon respectively. [5]

iii) Derive an expression for the output voltage of the piezoelectric bridge as a function of the induced strain, and hence calculate the sensitivity (output voltage per unit acceleration) of the device when the bridge is connected to a 5 V supply. The density of silicon is 2330 kgm^{-3}, and you may assume a value of 160 GPa for Young's modulus and values of $\pi_L = 72 \times 10^{-11}$ Pa^{-1} and $\pi_T = -66 \times 10^{-11}$ Pa^{-1} for the reduced piezoelectric coefficients. [5]

iv) Solve the bending equation for either cantilever to obtain its deflection profile. Hence determine the deflection at the far end of the proof mass when the applied acceleration is 100 ms^{-2}. [10]

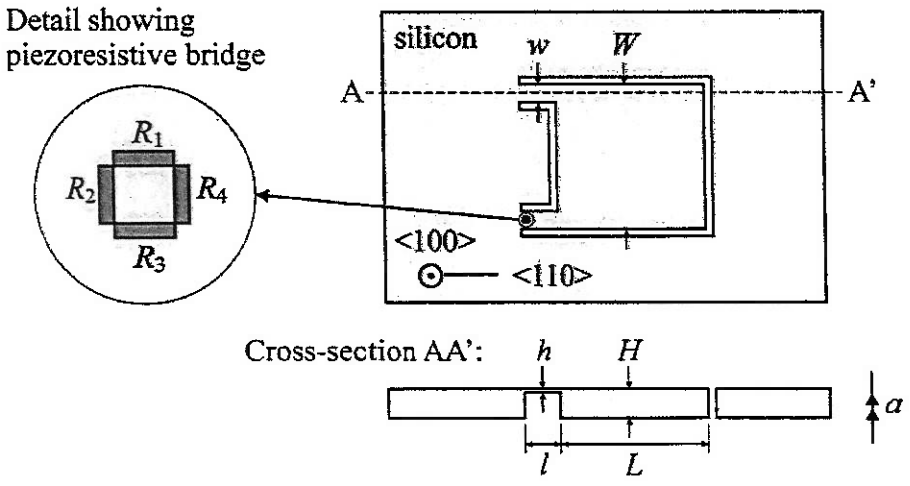


Figure 4.1

5. a) Figure 5.1 shows the electromechanical model for an electrostatic actuator comprising a moveable plate suspended over a fixed plate covered by a dielectric layer. Assuming the dielectric layer is of negligible thickness, derive an expression for the total force acting on the moveable plate. [4]

Sketch the variation of the total force with the gap g for several values of V and explain, with the aid of your sketch, the origin of *snap-down* instability. [6]

b) State the conditions that apply at the point of snap-down. Hence show that, if the dielectric layer is neglected, the snap-down voltage is expected to be:

$$V_P = \sqrt{\frac{8kg_0^3}{27\varepsilon_0 A}} \qquad [8]$$

c) Figure 5.2 shows an electrostatically actuated, capacitive shunt RF MEMS switch designed for low actuation voltage. It consists of a gold plate suspended over a transmission line by a simple hammock suspension. The suspension beams are each 250 μm long and 4 μm wide, and the gold layer is 2 μm thick. The total plate area is 72,500 μm^2 and you may assume all of this area contributes to the actuation force. The dielectric spacer is a thin layer of silicon nitride, and the initial gap between the plate and the dielectric is 4 μm.

Estimate the snap-down voltage of the switch, stating any assumptions needed for your calculation. You should assume $E = 80$ GPa for gold. [6]

d) A rough estimate for the switching time at larger actuation voltages can be obtained by assuming the moveable plate is acted on only by the electrostatic force, and that this force remains constant as the gap decreases. Using this approximation derive an estimate for the switching time when $V = 20$ V. The density of gold is 19,320 kgm^{-3}. [6]

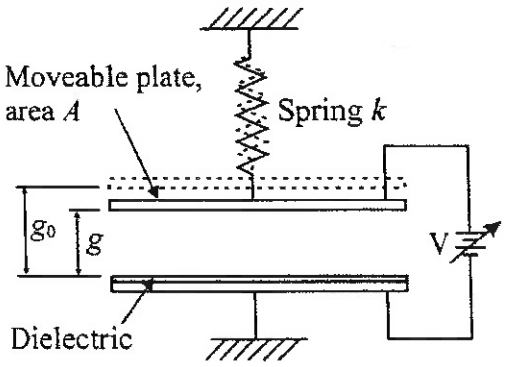


Figure 5.1

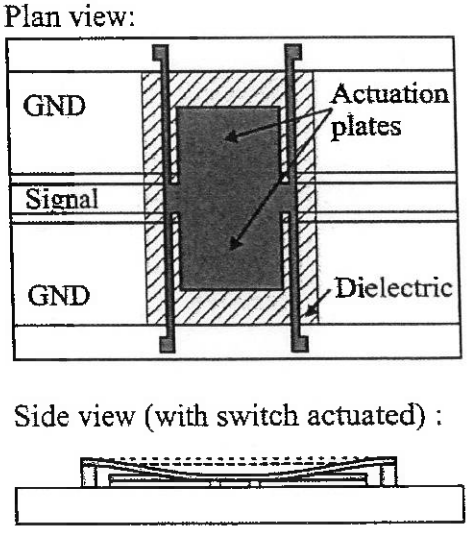


Figure 5.2

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